# Kenneth EZUKWOKE, Ph.D


## Personal Information

Kenneth EZUKWOKE, Ph.D
Research & Development Engineer - Applied Math./GenAI, EMSE
+33(0)767962527 Saint-Étienne, France
ifeanyi.ezukwoke@emse.fr kennedyczar.github.io
GoogleScholar Researchgate github.com/kennedyCzar
www.linkedin.com/in/kenneth-ezukwoke-4497b1155/

## Professional Objective

I am currently a Data Scientist at Datategy working on papAI product development. This comes after obtaining my Ph.D in Applied Mathematics-GenAI and industrial engineering from École Nationale Supérieure des Mines de Saint-Étienne, working on Artificial Intelligence for Failure Analysis in Semiconductor Industry 4.0- an industrial collaboration with STMicroelectronics, Bosch, and Infineon AG under the FA4.0 EURIPIDES2-PENTA project. My research is focused on improving Generative Artificial intelligence via Bayesian Inference and Causal Large Language Models (LLMs), for decision-making during Failure Root Cause Analysis (FRCA). Advised by Professor Mireille Batton-Hubert, Professor Xavier Boucher (both from EMSE) and Pascal Gounet and Jerome Adrian (both Physical Failure Analyst Engineers, STMicroelectronics).

## Awards

**Top 10 of 100+ teams**, AIRBUS Anomaly detection challenge, 2020
*Top 10 of 100+ teams across France in the **AIRBUS** anomaly detection challenge. Modeling using Variational Autoencoder and Desity based clustering.*

**Summer Grant**, ACM SIGCHI Summer Grant, 2019
***ACM SIGCHI** Summer Grant Recipient ACM Summer Grant. Summer school held in Barcelona, Spain, focused on user modeling and personalization in urban computing.*

**STUDENT RESEARCH GRANT**, UCL QATAR, 2018-2019
*Student research grant awarded by UCL QATAR on data management system for UCL Qatar student-led event.*

**GOOGLE developer challenge scholarsip**, UDACITY Europe, 2017
*Google Europe Udacity Scholarship recipient.*

**FIRST CLASS DISTINCTION**, VISTAS, 2017
*Annual convocation ceremony Vels University.*

## Education

**Ph.D in Applied Math./Generative AI** — Saint-Étienne, France
École des Mines de Saint-Étienne (EMSE) — *Oct. 2020 - Sept. 2023*

- PhD candidate in Applied Math. / A.I, Researching GenAI for Failure Analysis FA4.0 EU Project
- An EU collaboration between EMSE, STMicroelectronics and Infineon Technologies AG with the objective of developing an AI-based solution for fault analysis in Microelectronics manufacturing industries of the future.

**M.Sc. Machine Learning and Data Mining [Bien (Good) ]** — Saint-Étienne, France
Université Jean Monnet (UJM) - Membre de l'Université de Lyon — *Sept. 2019 - Aug. 2020*

- Dual masters degree [MLDM] provides an original scientific position in Europe on problems related to machine learning, big data, pattern recognition, advance machine learning and advance AI.

**B.Sc. Computer Application [Distinction ]** — Chennai, India
Vels Institute of Science Technology and Advance Studies (VISTAS) — *June 2014 - May 2017*

- Bachelors focusing on Statistics, Calculus, Algorithms and Data structure, software/Application development. Final Thesis on designing a CRM application for student monitoring.

## Online Education

**Foundations of machine learning - MEHRYAR MOHRI**
New York University (Courant Institute and Google Research online) — *2018*

- Regression, Convex optimization, kernel methods, Multi-class classification, Boosting, Reinforcement learning, Online learning, learning with finite and infinite hypothesis.

**Machine learning for financial trading**
Georgia Tech University — *2017*

- Deep learning algorithms, Machine Learning, Big data analysis and Stock prediction algorithms.

## Publications

K. Ezukwoke, H. Toubakh, A. Hoayek, M. Batton-Hubert, X. Boucher and P. Gounet, "Intelligent Fault Analysis Decision Flow in Semiconductor Industry 4.0 Using Natural Language Processing with Deep Clustering," *2021 IEEE 17th International Conference on Automation Science and Engineering (**CASE**)*, 2021, pp. 429-436, doi : 10.1109/CASE49439.2021.9551492 .

K. Ezukwoke, A. Hoayek, M. Batton-Hubert and X. Boucher, "GCVAE : Generalized-Controllable Variational AutoEncoder," *arXiv*, 2022, doi : 10.48550/ARXIV.2206.04225.

Z. Wang, K. Ezukwoke, A. Hoayek, M. Batton-Hubert and X. Boucher, "NLP based on GCVAE for intelligent Fault Analysis in Semiconductor industry," *IEEE International Conference on Emerging Technologies and Factory Automation (**ETFA-2022**)*, doi : 10.1109/ETFA52439.2022.9921524.

Ezukwoke, K., Hoayek, A., Batton-Hubert, M., Boucher, X., Gounet, P., and Adrian, J. (2022). Leveraging Pre-trained Models for Failure Analysis Triplets Generation. *arXiv*, doi : 10.48550/arXiv.2210.17497.

A. Rammal, K. Ezukwoke, A. Hoayek and M. Batton-Hubert, "Root cause prediction for failures in semiconductor industry, a genetic algorithm–machine learning approach," Scientific Reports 13(1), 4934, doi : 10.1038/s41598-023-30769-8.

## Accepted

K. Ezukwoke, A. Hoayek, M. Batton-Hubert and X. Boucher, "GCVAE : Generalized-Controllable Variational AutoEncoder," ***BELIEF 2022. Poster.***

K. Ezukwoke, A. Hoayek, M. Batton-Hubert and X. Boucher, "$\beta$- Variational AutoEncoder and Gaussian Mixture Model for Fault Analysis Decision Flow in Semiconductor Industry 4.0," *European Network for Business and Industriall Statistics (**ENBIS-21**)*. **Poster**.

K. Ezukwoke, A. Hoayek, M. Batton-Hubert, X. Boucher and P. Gounet, "Big GCVAE : Decision-Making with Adaptive Transformer Model for Failure Root Cause Analysis in Semiconductor Industry *Journal of Intelligent Manufacturing (**JIM**)*.

## Under peer-review

K. Ezukwoke, A. Hoayek, M. Batton-Hubert and X. Boucher, "GCVAE : Generalized-Controllable Variational AutoEncoder," *Journal of Machine Learning Research (JMLR)*. **Accepted**.

K. Ezukwoke, A. Hoayek, M. Batton-Hubert, X. Boucher and P. Gounet, "FAGPT : Leveraging Pretrained Models for Failure Analysis Triplet Generation." *Journal of Intelligent Manufacturing (JIM)*. **Accepted**.

## Professional Experiences

**Data Scientist- Generative AI** — Paris, France
Datategy — *Mar. 2024 - Present*

- Generative artificial intelligence technology research and development for papAI.

**Artificial Intelligence for Fault Analysis (FA4.0 EU Project)** — Grenoble, France
STMicroelectronics — *Oct. 2020 - Sept. 2023*

- Designing an A.I solution based on probabilistic graphical model for decision-making during failure root cause analysis observed during production of microelectronics.

**Machine Learning for Anomaly Detection on Complex Chemical Reactors** — Lyon, France
IFP Energies nouvelles — *Feb. 2020 - Aug. 2020*

- Analysis of complex sensor time-series to detect anomalies during process monitoring of a pilot plant. Develop an a posteriori automated machine learning algorithm for early anomaly detection. Analysis of complex sensors, trends and potential anomalies.

**DATA ANALYST/PYTHON DEVELOPER** — Wurzburg, Germany
Wölfel Engineering GmBH — *April 2018 - Aug. 2018*

- Algorithm design and implementation for handling Wind turbine data from Offshore Windpark.
- Data analysis and visualization using white label technology for in-house usage.

**DATA ANALYST/PYTHON DEVELOPER** — Chennai, India
Arun Cyber Technologies Ltd. (Startup Incubation Lab) — *Oct. 2016 - April 2017*

- Data analysis for detecting cyber threat, intrusion and maintenance.

**C# Software Developer** — Chennai, India
Vels Institute of Science, Technology Advanced Studies — *2015 - 2016*

- C# Software Developer for digital software department.
- Customer Relationship Management (CRM) software developer for Parent Student Monitor.
- University Database management system maintenance.
- SQL programming for database management.

## Service

**Reviewer :** *WiML, IEEE CASE, EMNLP, AISTATS, ACL.*

## Affiliations

**Organizations :** *CNRS UMR-6158 LIMOS, Blacks in AI, Masakhane.*

## Volunteering

**The 10th International Conference on Learning Representations** — Virtual
**ICLR**. Sponsors : Meta, DeepMind, Google Research, Byte Dance, Microsoft — *Apr. 25-29, 2022*

**The 25th International Conference on Artificial Intelligence and Statistics** — USA. [Virtual]
**AISTATS**. Sponsors : G-Research, Google DeepMind, Facebook AI — *March 28-30, 2022*

**IEEE International Conference on Automation Science and Engineering** — Lyon, France
**IEEE CASE**. Sponsors : IEEE, EMSE, INSA, IMT, LIMOS — *Aug. 23-27, 2021*

**Mentor on AI for Coral Reef Conservation in the Vamizi Island** — Mozambique
Sponsors : Sida, K4A, IDRC-CRDI Canada, UNESCO-IRCAI — *Oct. 2020 - 2021*

**The 24th International Conference on Artificial Intelligence and Statistics** — USA. [Virtual]
**AISTATS**. Sponsors : G-Research, Google DeepMind, Facebook AI — *April 13-15, 2021*

## Skills

**Operating systems :** Linux, Windows.
**Programming languages :** Python, Bash, R, MATLAB.
**Office softwares :** LaTeX, Libre Office, Microsoft Office, Open Office.
**Scientific packages :** Pandas, Numpy, Sklearn, Tensorflow, Keras, Pytorch.
**Web applications :** Django, Flask, Plotly Dash, Bokeh, Git.
**Optimization :** Algorithmic and convex optimization.
**Languages :** English, French.
**Independent projects :** Kernel methods, Cheese-coin (blockchain), Graph mining, Forecasting 1.0.

## Invited talks

**Plate-Forme Intelligence Artificielle (PFIA-22)** — Saint-Étienne, France
Presentation : Artificial intelligence for Fault Analysis in Semiconductor Industry
Highlight : GCVAE with Transformers for structured data generation.

## Teaching experience

**Big data clustering** — Saint-Étienne, France
École des Mines de Saint-Étienne (EMSE) — January, 2022

**Introduction to Natural Language Processing (NLP)** — Saint-Étienne, France
École des Mines de Saint-Étienne (EMSE) — January, 2022

**Deep Learning practical class** — Saint-Étienne, France
École des Mines de Saint-Étienne (EMSE) — 2021

Last updated, **6 mars 2024**.